University Of Engineering and Technology, Lahore
Computer Engineering Department

Lab Evaluation:

Course Name: Object Oriented Programming	**Course Code:** CMPE 221L
Assignment Type: Lab	**Dated:**
Semester: 2nd	**Session:** CYS 2026 Spring
Lab/Project/Assignment #: 5	**CLOs to be covered:** CLO1
Lab Title: Constructors & Modifiers	**Teacher Name:** Hafiz Muhammad Mubashar Imran

CLO 1	Implement abstraction and encapsulation to develop reusable classes for objects of real-world problems					
Levels (Marks)	**Level 1**	**Level 2**	**Level 3**	**Level 4**	**Level 5**	**Level 6**
(10)						
					Total	**/10**

Rubrics for Current Lab Evaluation

Scale	**Marks**	**Level**	**Rubric**
Excellent	**9-10**	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	**7-8**	L2	Submitted the lab tasks but have good understanding
Good	**5-6**	L3	Submitted the lab tasks but have weak understanding.
Basic	**3-4**	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	**1-2**	L5	Submitted only one lab task.
Not Acceptable	**0**	L6	Did not attempt

LAB No. 5

Lab Goals/Objectives:

- Basic OO programming in java
- Objects and Classes
- Constructor
- Static and Final

1. Design and implement a class dayType that implements the day of the week in a program. The class dayType
should store the day, such as Sun for Sunday. The program should be able to perform the following operations
on an object of type dayType:
a. Set the day.
b. Print the day.
c. Return the day.
d. Return the next day.
e. Return the previous day.
f. Calculate and return the day by adding certain days to the current day.
For example, if the current day is Monday and we add 4 days, the day to be returned is Friday. Similarly, if today is Tuesday and we add 13 days, the day to be returned is Monday.
Code:

```
import java.util.Scanner;
class dayType {
    private String[] days = {"Sun", "Mon", "Tue", "Wed", "Thu", "Fri", "Sat"};
    private int currentDay;
    public dayType() {
        currentDay = 0;
    }
    public void setDay(String day) {
        for (int i = 0; i < 7; i++) {
            if (days[i].equalsIgnoreCase(day)) {
                currentDay = i;
            }
        }
    }
    public void printDay() {
        System.out.println("The day is " + days[currentDay]);
    }
    public String getDay() {
        return days[currentDay];
    }
    public String nextDay() {
        int next = currentDay + 1;
        if (next == 7) {
            next = 0;
        }
        return days[next];
    }
    public String previousDay() {
        int previous = currentDay - 1;
        if (previous == -1) {
            previous = 6;
        }
        return days[previous];
    }
    public String addDays(int numberOfDays) {
        int newDay = (currentDay + numberOfDays) % 7;
        if (newDay < 0) {
            newDay = newDay + 7;
        }
        return days[newDay];
    }
}
public class Task01 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
```

```
        dayType today = new dayType();
        System.out.print("Enter the day (Sun, Mon, Tue, Wed, Thu, Fri, Sat): ");
        String day = input.next();
        today.setDay(day);
        today.printDay();
        System.out.println("Current day: " + today.getDay());
        System.out.println("Next day: " + today.nextDay());
        System.out.println("Previous day: " + today.previousDay());
        System.out.print("Enter number of days to add: ");
        int numberOfDays = input.nextInt();
        System.out.println("Day after adding " + numberOfDays + " days: " +
today.addDays(numberOfDays));
    }
}
```

Output:

```
/usr/lib/jvm/default-java/bin/java -javaagent:/snap/intellij-idea/13
Enter the day (Sun, Mon, Tue, Wed, Thu, Fri, Sat): fri
The day is Fri
Current day: Fri
Next day: Sat
Previous day: Thu
Enter number of days to add: 4
Day after adding 4 days: Tue

Process finished with exit code 0
```

2. Walls has decided that they are going to sell their ice cream in 7 different flavors namely chocolate, vanilla,
strawberry, mango, tutti fruit, almond crunch, and coffee.
Create a Walls class with flavor, number of scoops, cone type, and price fields.
If no arguments are supplied, the flavor defaults to "Vanilla," the number of scoops defaults to 1, and the
cone type defaults to "Sugar."
The constructor calculates the price based on Rs. 75 per scoop, with an additional Rs. 40 for a waffle cone.
Note: Use a Switch/If statement to ask the user flavor, scoop, and cone type in the main() function.
Code:

```
import java.util.Scanner;
class Walls {
    private String flavor;
    private int scoops;
    private String coneType;
    private int price;
    public Walls() {
        flavor = "Vanilla";
        scoops = 1;
        coneType = "Sugar";
        calculatePrice();
    }
    public Walls(String flavor, int scoops, String coneType) {
        this.flavor = flavor;
        this.scoops = scoops;
        this.coneType = coneType;
```

```
        calculatePrice();
    }
    private void calculatePrice() {
        price = scoops * 75;
        if (coneType.equals("Waffle")) {
            price = price + 40;
        }
    }
    public void display() {
        System.out.println("Flavor: " + flavor);
        System.out.println("Scoops: " + scoops);
        System.out.println("Cone type: " + coneType);
        System.out.println("Price: Rs. " + price);
    }
}
public class Task02 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.println("Default order:");
        Walls defaultOrder = new Walls();
        defaultOrder.display();
        System.out.println();
        System.out.println("Flavors:");
        System.out.println("1. Chocolate");
        System.out.println("2. Vanilla");
        System.out.println("3. Strawberry");
        System.out.println("4. Mango");
        System.out.println("5. Tutti Fruit");
        System.out.println("6. Almond Crunch");
        System.out.println("7. Coffee");
        System.out.print("Choose a flavor (1-7): ");
        int flavorChoice = input.nextInt();
        String flavor;
        switch (flavorChoice) {
            case 1:
                flavor = "Chocolate";
                break;
            case 2:
                flavor = "Vanilla";
                break;
            case 3:
                flavor = "Strawberry";
                break;
            case 4:
                flavor = "Mango";
                break;
            case 5:
                flavor = "Tutti Fruit";
                break;
            case 6:
                flavor = "Almond Crunch";
                break;
            case 7:
                flavor = "Coffee";
                break;
            default:
                System.out.println("Invalid choice, Vanilla selected.");
                flavor = "Vanilla";
        }
        System.out.print("Enter number of scoops: ");
        int scoops = input.nextInt();
        System.out.println("Cone types:");
```

```
        System.out.println("1. Sugar");
        System.out.println("2. Waffle");
        System.out.print("Choose a cone type (1-2): ");
        int coneChoice = input.nextInt();
        String coneType;
        if (coneChoice == 2) {
            coneType = "Waffle";
        } else {
            coneType = "Sugar";
        }
        System.out.println();
        System.out.println("Your order:");
        Walls order = new Walls(flavor, scoops, coneType);
        order.display();
    }
}
```

Output:

```
Flavor: Vanilla
Scoops: 1
Cone type: Sugar
Price: Rs. 75

Flavors:
1. Chocolate
2. Vanilla
3. Strawberry
4. Mango
5. Tutti Fruit
6. Almond Crunch
7. Coffee
Choose a flavor (1-7): 1
Enter number of scoops: 3
Cone types:
1. Sugar
2. Waffle
Choose a cone type (1-2): 2

Your order:
Flavor: Chocolate
Scoops: 3
Cone type: Waffle
Price: Rs. 265
```

3. Create a class by the named Product. Create the following private members name(String), price(int) and
quantity(int).
• Define a parameter-less and a parameterized constructor.
•Define the following member functions.

• void getData()
• static void counter()
• static int countProduct()
•The class would contain a static int ‘totalProduct’, having initial value equal to zero.
•Each time an instance of the product class is created, the ‘counter ()’ should be called. The ‘counter ()’
increments the static member by 1 value. Suppose I create 5 products, when I execute the following code it should display 5.
Code:

```
import java.util.Scanner;
class Product {
    private String name;
    private int price;
    private int quantity;
    private static int totalProduct = 0;
    private static Scanner input = new Scanner(System.in);
    public Product() {
        name = "Unknown";
        price = 0;
        quantity = 0;
        counter();
    }
    public Product(String name, int price, int quantity) {
        this.name = name;
        this.price = price;
        this.quantity = quantity;
        counter();
    }
    public void getData() {
        System.out.print("Enter product name: ");
        name = input.next();
        System.out.print("Enter product price: ");
        price = input.nextInt();
        System.out.print("Enter product quantity: ");
        quantity = input.nextInt();
    }
    public void display() {
        System.out.println("Name: " + name + ", Price: " + price + ", Quantity: " + quantity);
    }
    public static void counter() {
        totalProduct++;
    }
    public static int countProduct() {
        return totalProduct;
    }
}
public class Task03 {
    public static void main(String[] args) {
        Product p1 = new Product("Pen", 20, 100);
        Product p2 = new Product("Notebook", 150, 50);
        Product p3 = new Product("Marker", 60, 30);
        Product p4 = new Product();
        Product p5 = new Product();
        System.out.println("Enter data for product 4:");
        p4.getData();
        System.out.println();
        p1.display();
        p2.display();
        p3.display();
        p4.display();
```

```
        p5.display();
        System.out.println();
        System.out.println("Total products: " + Product.countProduct());
    }
}
```

Output:

```
Enter data for product 4:
Enter product name: pen
Enter product price: 35
Enter product quantity: 4

Name: Pen, Price: 20, Quantity: 100
Name: Notebook, Price: 150, Quantity: 50
Name: Marker, Price: 60, Quantity: 30
Name: pen, Price: 35, Quantity: 4
Name: Unknown, Price: 0, Quantity: 0

Total products: 5

Process finished with exit code 0
```